ORIGINAL ARTICLE

Comparative Effect of Propolis of Honey Bee and Some Herbal Extracts on *Candida Albicans*

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ABSTRACT **Objective**: To determine the effect of propolis on *Candida albicans* and to compare it with the effects of some other herbal extracts and antibiotics on this pathogenic fungi. **Methods**: The extracts of propolis, *Thymus vulgaris*, *Caryophillium aromaticus*, *Echinophora platyloba*, *Allium cepa* and *Cinnamomum zeylanicum* were prepared and the antifungi effects of the extracts were examined on *Candida albicans* ATCC10231 using disc-diffusion assay and micro-broth dilution. The minimum fungicidal concentration (MFC) and minimum inhibitory concentrations (MIC) as well as inhibition zone were evaluated and the anti fungi effects of herbal extracts were compared with amphotricin B and nystatin at the times of 24, 48 and 72 h. Data analysis was performed using *t* test. **Results**: Obtained results showed that propolis extract with MIC_{90} and MFC equal to 39 and 65 μg/mL, respectively, possess the highest antifungal activity when compared with other studied extracts. The extracts of *Allium cepa* and T*hymus vulgaris*, with MFC of 169 and 137 μg/mL, respectively, showed the lowest effects on the fungi. Also nystatin and amphotricin B yielded better effects on the tested fungi compared with the effects of all studied extracts on *Candida albicans*. **Conclusions**: Propolis extract is effective in controlling *Candida albicans*. However, the issue requires further investigation on samples in animals and performing toxicological examinations.

KEYWORS *Candida albicans*, propolis, herbal extract, antibiotics

Candida genus has several species but more pathogenic species in human are *Candida albicans*. It has been the fourth cause of blood infections in hospitalized patients.(1,2) The opportunistic fungi can cause many clinical complications in human such as thrush, vaginitis, skin infections, brain abcess, endocarditis, meningitis, arthritis and pyelonephritis. *Candida albicans* is found as normal flora in mucosal surface of the oral cavity, gastrointestinal tract and vagina. The fungi is colonized at mucosal surfaces and therefore more endogenous infections occur in this area.(3,4) Most people with healthy immune defense mechanisms, can control growth and spread of this opportunistic fungi, but under certain conditions with predisposing factors such as diabetes, immune deficiency and extensive use of antibiotics create an opportunity to shaping pathogen *Candida albicans* in the patients and cause systemic and fatal illness.(5-7) Drug resistance of this fungi, increase of drug usage doses and problems of side effects, have led to discovering new medicinal plant and natural materials with fewer side effects.(5,7,8) This study investigated the effects of some medicinal plants on *Candida albicans*. These plants are introduced below.

Thymus vulgaris (Thyme) is a member of the *Lamiaceae* family and variant of this plant is growing in mountains of Iran. In traditional medicine in Iran and some other countries, this plant is used for the treatment applications. Chemical constituents of Thyme are: essential oils (generally thymol and carvacrol), flavonoids, tannins and triterpenes.(9,10) The essential oil thyme may be suggested for food preservative because of strong antifungal and anti-aflatoxigenic efficacy.(11) Strong antifungal activity of thymol and carvacrol were reported in publications.(11-13) *Thymus vulgaris* thus consists of component that has antibacterial activity (against gram negative and gram positive bacteria).(14)

Zingiberofficinale is a member of *zingiberaceae* family. Its extract has several biological properties such as antiinflammatory, antioxidant, hepatoprotective, antitumoral, and antimicrobial activities.(15) Anti-microbial activity of this plant is referred to antimicrobial substances such aszingiberol,

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DOI: 10.1007/s11655-015-2074-9

zingiberin and bisabolene.(16) The antimicrobial activity of solvent extract of *Zingiberofficinale* was tested against multi drug resistant bacteria and fungi such as *Escherichia coli* and *Candida albicans*.(17) This plant has antimicrobial activity against *staphylococcus areuss*, *salmonella* sp and *Candida albicans* and *k. pneumonia*.(16) Antifungal activity of *Zingiber officinale* against Clotrimazole-resistant *C. albicans* had also been reported.(18)

Cinnamomum verum is a member of *Lauraceous* family. Inner bark of this plant is used as spice. It thus acts as an antimicrobial constituent of cinnamon and consists of 0.5% to 2.0% volatile oil in which generally eugenol and cinnamic aldehyde are active components. It was found that cinnamon has antibacterial activity against gram negative and gram positive bacteria.(19) Antibacterial activity of its essential oil has been reported against drug-resistant *Staphylococcus*, coagulase negative, *S. aureus*, *S. pneumonia* and *Enterococcus spp*.(20) Thus antifungal cinnamon activity against *Candida spp* such as *C. albicans* was reported.(21) *Echinophoraplatyloba* is a member of *Apiaceae* family. This plant has been applied in Iranian traditional medicine. It contains saponin, alkaloid and flavonoid.(22)

It was reported that *E. platyloba* has an activity against *Candida albicans*.(23) Antibacterial activity of this plant had also reported against *Staphilococusaureus* and *Pseudomonas aeroginosa*.(22) *Foeniculum vulgare* is a member of *Apiaceace* family. Its oil antimicrobial activity is reported against *Aspergillusfumigatus* and *A. niger*.(24) *Allium cepa* (onion) is a member of *Alliacae* family in traditional medicine, and it has been used for the treatment of intestinal infections.(25) Antibacterial (for example against *Vibrio cholera*), antiviral, antiparasitic, antifungal (for example against *Candida albicans*)(26) and antioxidant activity of onion has been reported.(27) The clove oil of *Eugenia caryophyllata* (*Syzigium aromaticum L. Myrtaceae*) has medicinal applications. Eugenol (the main component of clove oil(28) has been reported to possess useful antiseptic and numbing effects.(29) Antifungal activity of *E. caryophyllata* essential oils against food borne fungal has also been reported.(30) The oils have antifungal activity against *Candida albicans* and were reported that this activity may depend on free phenol hydroxyl group and aromatic rings.(31)

METHODS

Propolis

Bee propolis is one of the valuable products in the hive. This material consists of flavonoids and other components which show antimicrobial activity. Antibacterial and antifungal properties of propolis was reported in many studies.(15,32-34)

Microbial Strain

In this study, standard strain of the *Candida albicans* ATCC10231 was prepared from Teb Research Institute in Esfahan, Iran. It was in the form of lyophilized powder and prepared on sabouraud dextrose agar (SDA). The strain was put at temperature 37 ℃ for 24 h. After incubation and growth of *Candida albicans*, 500 μL phosphate buffered saline (PBS), with 7.2 in pH, was put in a 1 mL micro-tube. Then by the use of sterile loop a few of the grown colony on SDA was transferred to PBS. After mixing the fungi in buffer by Neubauer lam, some fungi cells were counted under a microscope and a suspension with cellular density of 1×10^6 CFU/mL was prepared from the cells.

Preparing Extract

In this study, maceration method was used to prepare the extract by methanol solvent. Plants used in the test were *Z officinale*, *T vulgaris*, *C verum*, *F vulgare* and *E platyloba*. These plants were powder by a hummer mill and passed from mesh with 80 and 100 size. Thirty grams of each plant powder were dissolved in 100 mL of pure methanol 80%. In order to have a better extraction, solution was gently stirred with magnetic stirring into a sterile dark glass for 48 h. Next, the liquid part of each extract was separated by paper filtration Whatman No 42 from the solid part. For extract concentration, the solvent was put in rotary (model Laborata 4000, manufactured by Heidolph Company, Germany) at temperatures around 46–60 ℃. Extraction procedure was repeated three times and then the obtained material was weighed and poured into a frosted glass and was kept in refrigerator until usage. For preparation of propolis extract, 10 g of bee propolis sample was weighed and then 100 mL of 96% ethanol was added to it. Mixture was kept in a warm and shadowy place for 1 to 2 weeks. After this period, the mixture was filtered and placed in a refrigerator at the temperature of 1 to 4 ℃ for 1 day. The solution was filtered again and then was kept in a dark and impenetrable container. The remaining alcohol in suspension was completely separated by Soxhlet extractor, therefore pure alcoholic extract was obtained.(15)

Disk Method

For evaluation of antimicrobial activity of the extracts of propolis, *Z. officinale*, *T. vulgaris*, *C. verum*, *F. vulgare* and *E. platyloba*, disc-diffusion assay were used. This method is the most common way for the evaluation of major antimicrobial substances and is known as the Kirby-Bauer test.(35) In this way the disks first were soaked in extract solution and then used. Concentrations of 0.1%, 0.312%, 1.25%, 2.5%, 5%, 10%, 20% and 30% of alcoholic extract from bee propolis, *Z. officinale*, *T. vulgaris*, *C. verum*, *F. vulgare* and *E. platyloba* were used. *Candida albicans* was prepared 48 h before the experiment. Then, 2 to 3 colonies of *Candida albicans* were added to sterilized physiologic serum and its darkness was determined to be 0.5 McFarland (1×10^6 CFU/mL) which was prepared from the suspension on SDA. Then blanch discs with a diameter of 6.5 mm (manufactured by Diagnostic Mass, England) were soaked in sterile conditions for 0.5 h using sterile forceps and were put at certain distances from medium. Amphotericin B (10 mg) and Nystatin (50 mg) were used for positive controls. Petri dishes were incubated at 37 ℃ for 24 h. Then, inhibition zone around the discs was measured by Colis.(36)

Least Determination of Minimum Inhibitoy Concentration and Minimum Fungicical Concentration Using Dilution Method in Liquid Culture

In this experiment 96-well plate was used. In every well, 200 μL of culture medium sabouraud dextrose broth and 10 μL of suspension with cell density 1×10^6 CFU/mL were added and different amount of extracts (1 to 200 μg/mL) were also added to every well. The 96-well plate was put in a shaking incubator (with 150 g) at 37 ℃ for 24 h. Then after incubation, the darkness of wells was evaluated using colony counting and fungi growth. The well with the lowest concentration was considered as minimum inhibitory concentration (MIC) of extracts. Also MIC_{50} (50% decrease in number of colonies compared to control group) and MIC_{90} (90% decrease in number of colonies compared to the control group) were determined. In order to evaluate and determine minimum fungicidal concentration (MFC), components from all wells were cultured on solid SDA so that 10 μL of suspension presented in every well were taken with sampler and transferred to solid SDA containing chloramphenicol and then it was incubated at 37 ℃ for 24 h. The lowest dilution of different extracts, which inhibited *Candida albicans* to grow on SDA was considered as MFC.

Statistical Analysis

All experiments were performed with 3 replicates. Data were analyzed using SPSS software (version 20) via One way ANOVA method and mean comparison was done through Tukey's HSD method.

RESULTS

Comparative Effect of Propolis and the Herbal Extracts Using Disk Method

Results reported no inhibitory growth effect caused by extract of *Z. officinale* and *F. vulgare* on *Candida albicans*. *C. zeylanicum* at the time of 24 h, with concentration 30%, was more effective than other treatments. Concentration 30% *E. platyloba*, 30% propolis and 20% *C. zeylanicum* showed a better effect respectively. At the time of 48 h, 30% *C. zeylanicum* had a significant performance compared to other treatments, and after which the effects of 30% *E. platyloba* and propolis were considerable. At the time of 72 h, the treatments *C. zeylanicum* and propolis both with concentration of 30% were better than other treatments, and after which 30% *E. platyloba* was considerable. So at all time points, the concentration 30% *C. zeylanicum* showed the best performance and 30% propolis, as time increasing, became more effective and at the time of 72 h, it had an effect similar to *C. zeylanicum* (Table 1).

Comparison of Best Treatments with Two Antibiotics

At different time points, best effects were compared between the two antibiotics (Figure 1). At all times, nystatin showed a better effect compared with amfoterisin B and with 30% *C. zeylanicum* and propolis ($P<0.01$). At the time of 24 and 48 h, the effect of Amfoterisin B was better than *C. zeylanicum* and propolis with concentration of 30% ($P<0.01$). But at the time of 72 h, this antibiotic did not show any meaningful effect toward *C. zeylanicum* and propolis with concentration of 30% ($P>0.05$). The antibiotics effects were much more effective than other

Table 1. Comparative Effect of Propolis and the Herbal Extracts Using Disk Method (mm)

Treatment	Concentration (%)	Inhibition zone		
		24 h	48 h	72 h
Propolis	2.5	0.54	2.32	5.11
	5	1.63	3.33	4.50
	10	3.56	6.42	7.30
	20	5.45	9.61	10.39
	30	7.30	14.24	15.29
T. vulgaris	2.5	—	1.44	1.49
	5	1.18	3.04	3.50
	10	2.50	4.49	5.44
	20	4.49	6.55	7.14
	30	5.47	9.18	10.38
C. aromaticus	2.5	0.32	0.43	0.43
	5	1.48	2.32	2.71
	10	2.42	6.47	6.24
	20	5.08	8.19	8.13
	30	5.26	12.05	12.10
E. platyloba	2.5	0.59	0.91	1.47
	5	1.56	2.50	3.12
	10	3.08	5.51	5.55
	20	5.71	8.38	8.57
	30	8.36	13.62	13.81
Allium cepa	2.5	—	0.63	0.59
	5	0.69	1.47	2.53
	10	1.56	4.26	4.30
	20	3.22	6.37	6.52
	30	4.63	8.20	8.55
C. zeylanicum	2.5	1.46	1.59	2.13
	5	2.01	4.42	4.30
	10	4.44	8.06	8.26
	20	7.37	11.56	12.52
	30	10.65	15.53	16.18
SEM		0.17	0.15	0.42
CV (%)		7.96	4.35	10.98
HSD ($P<0.05$)		0.90	0.79	2.23

Notes: SEM: standard error of the mean; CV: coefficient of variation; HSD: Tukey honestly significant difference test

treatments used in the experiment.

MIC and MFC of Propolis and the Herbal Extracts

The concentrations of extract of propolis, *T. vulgaris*, *C. aromaticus*, *E. platyloba*, *Allium cepa* and *Cinnamomum zeylanicum* were 30, 26, 31, 25, 27, 23 mg/mL, respectively. The results obtained from MIC showed that MIC_{50} and MIC_{90} of propolis extract and *C. zeylanicum* extract had the best effects on *Candida albicans* (Table 2).

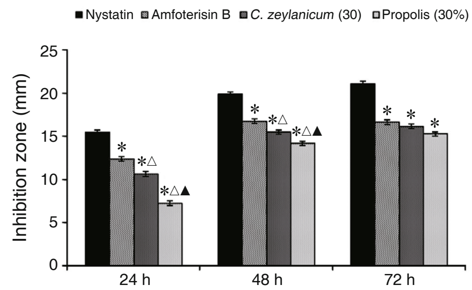

Figure 1. Comparison of Best Treatments with Two Antibiotics

Notes: $^{*}P<0.05$, compared with the nystatin; $^{\triangle}P<0.05$, compared with the amfoterisin; $^{\blacktriangle}P<0.05$, compared with the *C. zeylanicum* (30%)

Table 2. MIC and MFC of Propolis and the Herbal Extracts

No.	Extract	Concentration (mg/mL)	MIC (μg/mL)		MFC (μg/mL)
			MIC_{50}	MIC_{90}	
1	Propolis	30	21	39	65
2	*T. vulgaris*	26	68	86	137
3	*C. aromaticus*	31	48	67	95
4	*E. platyloba*	25	27	51	83
5	*Allium cepa*	27	75	93	169
6	*C. zeylanicum*	23	18	32	52

DISCUSSION

The word propolis is derived from the Greek words; pro ('in front of') and polis ('city' or 'community'), which means that this natural product associates to hive defense. Its protective properties come from components that contribute in propolis production. Many investigations proved the antibacterial activity of this substance.[37,38] The propolis of bees or plant resins is the product of worker bees. The bees gathered the plant resins in the hive with some changes to it, use it as sealant, polisher, disinfectant and to mummify the dead insects in hives. The propolis consists of resins, wax, volatile and pollens of flowers. Based on biochemical analysis, it was revealed that propolis consists of alcohols, aldehydes, phenolics, amino acids esters and fatty acids which are valuable in pharmaceutic industry.[15]

Researches showed that the propolis possess antimicrobial, antiunicellular, antiviruses and antifungal properties.[39] Considering the broad therapeutic spectrum of propolis, it can be used to heal some diseases. One of the most important compounds in propolis is phenolic which gives a considerable antimicrobial property to propolis. According to experiments all around the world, the propolis can control a broad spectrum of Gram negative and Gram positive bacteria. In relation to bacterial diseases, it is revealed that although propolis has a considerable effect on Gram positive bacteria, it has a few effect on Gram negative bacteria.[40] The antimicrobial properties of propolis may occur by direct action on microorganism or non-direct action by stimulation of immune system resulting in killing more microorganism. The *in vivo* and *in vitro* studies proved that propolis may activate macrophages, increase their antimicrobial activities and also stimulate the production of antibodies. The antimicrobial property of propolis is related to numerous quantities of phenolics in it which can be referred to some derivatives such as pinobanksin and pinocembrin. In addition to high percentage of phenolics, the antimicrobial property of propolis has also been referred to an activated part called caffeic acid phenethyl ester.[41-44] The fulfilled experiments about propolis in different places showed special phenolic compounds with different percentage in samples.[15] In an experiment the effect of propolis prepared from 4 regions of Turkey and also a propolis from Brazil were tested on 9 pathogenic bacteria exited in mouth. The results showed that the propolis prepared from different regions was effective on all 9 species of bacteria, but the propolis prepared from Kaza in Ankara city possessed a higher inhibitory effect compared with other propolis from different regions in Turkey and Brazil.[40]

In this study we tested the effect of propolis on *Candida albicans* at a special time intervals. At first the effect of propolis was compared with some herbal extracts obtained from *T. vulgaris*, *Zingiberofficinale*, *Cinnamomumverum*, *E. platyloba*, *Foeniculum vulgare*, *Allium cepa* (onion) and *Eugeniacaryophyllata* and then with two kinds of antibiotics Nystatin and Amfoterisin B. Concentration 30% of propolis, as time increasing, became more effective and at the time of 72 h had an effect equal to *C. zeylanicum*. At the time of 72 h, antibiotic Amfoterisin B did not show any meaningful difference compared with *C. zeylanicum* and propolis with concentration of 30%. In another study it was revealed that methanol extract of propolis 30% has an antimicrobial effect on oral microorganisms such as *Candida albicans*, *streptococcus mutans* and *actinobacillus actinomycetemcomitans*.[45,46]

Another study has assessed and compared the antifungal activities of two essentials oils extracted from *Ocimum gratissimum* and *C. zeylanicum* against fungi isolated from traditional cheese wagashi. Results indicate that the essential oils of *Ocimum gratissimum* and *C. zeylanicum* have an interesting antifungal activity. Meanwhile, essential of *Ocimum gratissimum* was the most active against all the tested strains. These oils especially *Ocimum gratissimum* essential oil could be used as natural antimicrobial agent in the fight against moulds species responsible for wagashi contamination. Furthermore, the biological activity of these oils is probably due to their prominent concentration in thymol for *Ocimum gratissimum* and *cinnamaldehyde* for *C. zeylanicum*.

From the obtained results, it is clear that the alcoholic extract of propolis can be used against the examining fungi *Candida albicans* in different concentrations. With regard to the fact that this substance shows antifungal activity, it is necessary to carry out studies on laboratory creatures in *in vivo* conditions, so that the advantages and the side effects of propolis can be examined carefully.

Conflict of Interest

No potential conflict of interest to this article was reported.

Authors Contribution

All the authors in this paper have equal proportion in conception and design, acquisition of data and analysis. They are also equal in drafting the article and revising it as well as final approval of the version to be published.

Acknowledgement

The authors would like to express their appreciation to Islamic Azad University of Khorasgan Branch for the valuable supports.

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(Received February 28, 2013)
Edited by WANG Wei-xia